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Dynamics of rotating triple systems: statistical escape theory versus numerical simulations

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ABSTRACT

Dynamical evolution of 120 000 equal-mass rotating triple systems is investigated. The system rotation is described by the parameter $w = -L_0^2 E_0 / G^2 m_0^5$, where G is the gravitational constant, m_0 is the mass of a body, and L_0 and E_0 are the angular momentum and the total energy of the triple system, respectively. We consider the values of $w = 0.005$, 0.1, 1, 2, 4 and 6. For each w, 20 000 triple systems are studied. The initial coordinates and velocities of the components are randomly chosen. The initial data are chosen in two different ways: the first one assumes a hierarchical structure initially and the second one does not. The evolution of each triple system is calculated until either the escape of one of the bodies occurs or the time exceeds 1000 mean crossing times of the system. The orbital parameters of the final binary and the escaper are recorded for each run. We compare the results of numerical simulations and predictions of a statistical escape theory. The statistical theory is based on the assumption of ergodicity, that is, the only information on the initial conditions remaining at the time of the escape of the third body is contained in the conserved total energy, total angular momentum and the mass values. The distributions of various quantities are derived from the allowable phase-space volumes. The distribution of binary energy agrees with earlier results by Heggie, with the angular momentum dependence of Mikkola & Valtonen being added. The eccentricities are distributed in general accordance with Monaghan's work, while the triple systems break up like in radioactive decay, as was previously found by Valtonen & Aarseth. The escape directions are preferentially perpendicular to the total angular momentum vector; the more so, the greater the angular momentum. The escape-angle distributions are derived from the statistical theory and are found to be in agreement with the numerical data. The relative orientations and magnitudes of the binary and third-body angular momenta are also explained based on the statistical theory.

Key words: stellar dynamics – celestial mechanics.

1 INTRODUCTION

Triple systems are common among stars and galaxies and therefore their dynamical evolution has been studied by many authors [see e.g. the reviews by Anosova & Orlov (1985), Valtonen (1988), Valtonen & Mikkola (1991) and Sterzik & Durisen (1998)].

In addition to numerical orbit integrations, the dynamical escape from a triple system may be studied in the framework of a statistical escape theory (Monaghan 1976a,b; Nash & Monaghan 1978). This approximate theory is based on the supposition that the phase trajectory of a triple system is quasi-ergodic in the region of strong interaction of the bodies (close triple approach). Then the probability of escape with definite final-state parameters (orbital elements of the final binary and the escaper) is proportional to the corresponding volume of the phase space in the centre-of-mass coordinate system. After the close triple approach, one may assume the absence of interactions between the escaper and the final binary.

Numerical simulations of rotating triple systems [see e.g. Anosova (1969), Standish (1972), Saslaw et al. (1974), Valtonen (1974), Anosova et al. (1984), Anosova & Orlov (1986) and Mikkola & Valtonen (1986)] have shown that an increase in the total angular momentum leads to a slow-down of dynamical evolution and a greater fraction of stable systems.

A comparison between the statistical escape theory and computer simulations (Monaghan 1976b; Nash & Monaghan 1978) shows a qualitative agreement for the distributions of final binary eccentricities and binding energies, as well as for escaper velocities. A trend

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towards orthogonality between the escape-velocity vector and the angular momentum vector (i.e. the angle θ between these vectors being close to 90°) is expected to become stronger when the angular momentum grows. This is in agreement with the numerical simulations by Valtonen (1974).

In the theory of Monaghan (1976b), the ensemble of three-body systems is assumed to possess a constant total energy E_0 and a constant angular momentum L_0. The volume of the available phase space is calculated under these restrictions, and it is used to calculate the statistical properties of the ensemble after the break-up of the three-body systems.

Promising as this statistical theory is, it also has shortcomings. The theory contains a free parameter, the radius of the sphere of interaction R, which enters into the final distributions. Even when the value of the parameter is optimized, the distributions of escape energy do not have quite the right shape, and the correct dependence on angular momentum is not reproduced. Moreover, it is not possible to carry out the calculations analytically, which makes it difficult to see how the theory could be adjusted for a better agreement with numerical experiments.

An alternative approach, introduced by Mikkola & Valtonen (1986), is to combine the analytical theory with numerical orbit calculations in a more straightforward theory. The theory was further elaborated by Valtonen & Karttunen (2005), and we will give a brief description of the theory following the latter authors. In this theory, the ensemble of triple systems possesses a common total energy E_0 but the total angular momentum is allowed to vary within reasonable limits. Alternatively, it may be required that the ensemble has a common value of total angular momentum L_0 but the total energy is allowed to vary within certain limits. The phase-space limitation arising from the second, non-constant parameter is introduced afterwards using numerical experiments and with the guidance of some basic principles. This leads to a simple description of the results of the experiments.

This work continues our previous study (Valtonen et al. 2003). We consider here dynamical evolution of rotating triple systems in 3D space. The distributions of some parameters describing the final states of break-up systems are constructed. A comparison with analytical theory is carried out.

2 MODEL

We consider dynamical evolution of triple systems with equal-mass components. Rotation is defined by parameter

$$w = \frac{-L_0^2 E_0}{G^2 m_0^5}, \tag{1}$$

where G is the gravitational constant, m_0 is the mass of a body, and L_0 and E_0 are the angular momentum and the total energy of triple system, respectively. The same parameter was used by Mikkola & Valtonen (1986).

In simulations, we use the following system of units: gravitational constant $G = 1$, masses of bodies $m_0 = m_a = m_b = m_s = 1$ and $E_0 = -1$.

Two different methods of generating initial conditions are used.

The first one is the same as in Mikkola & Valtonen (1986). Dynamical evolution is started from an elliptic encounter of a single body and a close binary. The binding energy of the outer binary, composed of the single body and the close binary, is small in comparison to the close binary energy. Thus, the triple system has a hierarchical structure initially.

The second method does not necessarily assume a hierarchy. The initial configuration is randomly chosen within so-called region D of the Agekian and Anosova map [see e.g. fig. 1 in Valtonen et al. (2003)]. In the Agekian–Anosova map, the positions of the two bodies with the greatest mutual separation are mapped at points: (−0.5, 0), the first body; and (+0.5, 0), the second body. The third body is placed somewhere within unit radius of the first body, and closer to the second body than to the first body. The region D so defined contains all possible configurations of triple systems (Agekian & Anosova 1967). Initial velocities are distributed isotropically when the virial ratio is randomly distributed within the interval (0, 1).

The quantity w (equation 1) is given one of the several fixed values, within the tolerance of 0.01 units. The values used are $w =$ 0.1, 1, 2, 4 and 6. For each w and each method of generating initial data, we study the evolution of 10 000 triple systems. Thus, the total number of runs is 100 000. Calculations are performed until the escape of one of the bodies occurs or until the time equals 1000 T_{cr}, where

$$T_{cr} = \frac{GM^{5/2}}{|2E_0|^{3/2}} \tag{2}$$

is mean crossing time. Here $M = m_a + m_b + m_s$ is the total mass of triple system. The length unit is the mean size

$$d = \frac{G(m_a m_b + m_a m_s + m_a m_s)}{|E_0|}\,. \tag{3}$$

The velocities are given in units of d/T_{cr}.

For a comparison, we also consider 20 000 triple systems at $w =$ 0.005 (with the tolerance of 0.0025 units). This case corresponds to a very small angular momentum. The limiting case of $w = 0$ is the plane one and will be considered elsewhere.

The simulations of hierarchical systems were performed using chain regularization (Mikkola & Aarseth 1993), as well as the numerical experiments with triple systems, which have initial configurations with region D, were made by three-body regularization (Aarseth & Zare 1974). The Bulirsch & Stoer (1966) integrator was used in both the cases. Both regularization schemes allow one to treat with close double and triple encounters without any essential loss of accuracy.

3 RESULTS OF SIMULATIONS

We have studied the following parameters of the systems after the escape.

(i) The number N of systems that break up within 1000 T_{cr}.
(ii) The lifetime T.
(iii) The semimajor axis a of the final binary.
(iv) The eccentricity e of the binary.
(v) The asymptotic relative velocity v_s of the escaper with respect to the centre of mass of the triple.
(vi) The angle θ between the angular momentum vector of the triple system and the velocity vector of the escaper.
(vii) The angle λ between the angular momentum vectors of the (inner) binary and the outer binary. The latter is formed by the centre of mass of the (inner) binary and the escaper.
(viii) The ratio β of the angular momenta of the (inner) binary and the outer binary.

The detailed analysis of these parameter distributions has been described by Valtonen et al. (2003). Here, we give a brief description of the main results. They are summarized in Table 1, where the medians of these parameters are shown for both methods of generating

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Table 1. Medians of escape parameters.

	w	0.005	0.1	1	2	4	6	6 ($T > 10\,T_{\rm cr}$)
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
N	H	9811	9814	9543	8700	4591	883	511
	D	9886	9839	9705	9447	7269	4692	618
T	H	8.2	16.5	26.7	36.9	60.3	160	430
	D	6.5	14.6	36.4	52.5	56.3	4.7	409
a	H	0.125	0.124	0.142	0.149	0.159	0.165	0.165
	D	0.071	0.106	0.136	0.143	0.152	0.130	0.164
e	H	0.823	0.758	0.736	0.754	0.756	0.651	0.636
	D	0.779	0.744	0.742	0.750	0.750	0.747	0.681
$v_{\rm s}$	H	1.23	1.23	0.90	0.73	0.45	0.23	0.23
	D	1.33	0.87	0.55	0.47	0.35	0.61	0.15
θ	H	66.8	71.7	82.8	84.6	86.3	87.3	87.3
	D	68.3	73.2	83.2	85.1	86.9	87.7	88.0
λ	H	167.3	121.9	90.1	81.9	59.5	36.0	35.6
	D	155.4	116.0	83.0	71.8	53.9	73.1	25.1
β	H	0.942	0.688	0.300	0.218	0.171	0.175	0.179
	D	1.717	0.728	0.281	0.205	0.150	0.111	0.148

initial data, noted as 'H' and 'D'. Here N is the number of runs with the escape of one body before $1000\,T_{\rm cr}$.

When the angular momentum of the triple system increases, the following trends are observed.

(i) The fraction of long-living systems with $T > 1000\,T_{\rm cr}$ grows;
(ii) the mean lifetime of the systems where three-body break-up has occurred increases;
(iii) the final binaries become on average softer;
(iv) the median eccentricity does not depend on w;
(v) the velocities of the escapers decrease on average;
(vi) as a rule, escapes are prograde and take place at nearly orthogonal direction to the angular momentum vector of the triple system; and
(vii) the fraction of the angular momentum which is carried away by the escaper increases, while the ratio of the (inner) binary orbital angular momentum to the angular momentum of the outer system decreases.

Let us note that the case of $w = 6$ differs from the general trends. This may be connected with fast escapes for which the ergodic hypothesis may not be valid. In the last column of Table 1, the results are indicated for the systems with lifetimes $T > 10\,T_{\rm cr}$ at $w = 6$. Those are in qualitative agreement with general trends.

Valtonen et al. (2003) present a short comparison between numerical simulations and Monaghan's statistical escape theory. Here, we will consider a modified statistical theory.

4 ANALYTICAL THEORY

4.1 Phase-space volume

In the Monaghan's theory, the three-body systems are studied after one of the bodies has escaped (Fig. 1). A body of mass $m_{\rm s}$ escapes

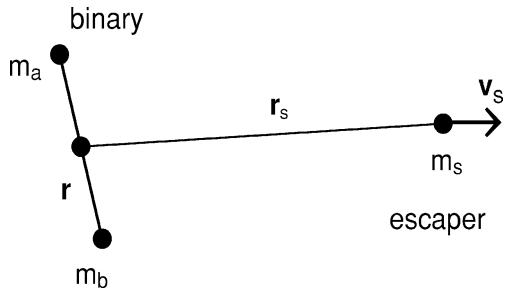


Figure 1. The basic configuration of three bodies after the body with mass $m_{\rm s}$ has escaped from the binary with component masses $m_{\rm a}$ and $m_{\rm b}$. The current speed of the escaper relative to the binary centre of mass is $v_{\rm s}$ and the current distance $r_{\rm s}$. The separation of the binary components is r at this time.

from the binary of component masses $m_{\rm a}$ and $m_{\rm b}$. The position vector of the third body relative to the barycentre of the binary is $\boldsymbol{r}_{\rm s}$ while the binary components are separated by the vector $\boldsymbol{r}$. We call the binary mass $m_{\rm B} = m_{\rm a} + m_{\rm b}$ and the total mass $M = m_{\rm B} + m_{\rm s}$, and the reduced masses $\mathcal{M} = m_{\rm a}m_{\rm b}/m_{\rm B}$ and $m = m_{\rm B}m_{\rm s}/M$. The potential energy of the escaper relative to the binary is approximately $V_{\rm s} = -Gm_{\rm B}m_{\rm s}/r_{\rm s}$ and the total energy is

$$E_0 = \frac{1}{2}m\dot{r}_{\rm s}^2 - G\frac{m_{\rm B}m_{\rm s}}{r_{\rm s}} + \frac{1}{2}\mathcal{M}\dot{r}^2 - G\frac{m_{\rm a}m_{\rm b}}{r} \tag{4}$$

or

$$E_0 = E_{\rm s} + E_{\rm B}, \tag{5}$$

where the energy is divided into the binary energy $E_{\rm B}$ and the energy of the escaper relative to the barycentre of the binary $E_{\rm s}$.

The density σ of escape configurations in the phase space per unit energy is obtained by integrating a δ-function over the phase-space coordinates $\boldsymbol{r}$, $\boldsymbol{r}_{\rm s}$, $\boldsymbol{p}$ and $\boldsymbol{p}_{\rm s}$, where $\boldsymbol{p}$ and $\boldsymbol{p}_{\rm s}$ are the momenta of the two relative motions. Thus,

$$\sigma = \int\int\int\int \delta\left(\frac{p_{\rm s}^2}{2m} + V_{\rm s} + E_{\rm B} - E_0\right) {\rm d}\boldsymbol{r}_{\rm s}\,{\rm d}\boldsymbol{p}_{\rm s}\,{\rm d}\boldsymbol{r}\,{\rm d}\boldsymbol{p}, \tag{6}$$

where we have put $E_{\rm s} = p_{\rm s}^2/2m + V_{\rm s}$.

Monaghan (1976a) integrates over $\boldsymbol{r}_{\rm s}$ and $\boldsymbol{p}_{\rm s}$ assuming that the escaper orbit is radially outward from the barycentre of the binary. We assume instead that the orbits of the escapers come from inside a 'loss cone' of base radius $7a$ where a is the semimajor axis of the binary (Fig. 2). The number seven is a practical choice based on numerical orbit calculations. The semimajor axis is related to the binary energy by

$$a = -\frac{Gm_{\rm a}m_{\rm b}}{2E_{\rm B}}. \tag{7}$$

The loss cone directions contain the fraction $\pi(7a)^2/(4\pi r_{\rm s}^2) = 12.25(a/r_{\rm s})^2$ of the whole sphere of radius $r_{\rm s}$ surrounding the escaper.

The integration over $\boldsymbol{r}_{\rm s}$ and $\boldsymbol{p}_{\rm s}$ is carried out as in Monaghan (1976a) with the result

$$\sigma = 98\sqrt{2}\pi^2(GMR)^{1/2}m^2(Gm_{\rm a}m_{\rm b})^2 \int\int \frac{{\rm d}\boldsymbol{r}\,{\rm d}\boldsymbol{p}}{|E_{\rm B}|^2}, \tag{8}$$

where R is the upper limit of the range of $r_{\rm s}$. The result is only approximate, but correct within 10 per cent when R is relatively small, say $R = 3a$.

In order to make direct comparisons with experiments, it is convenient to transform the remaining variables of integration and to

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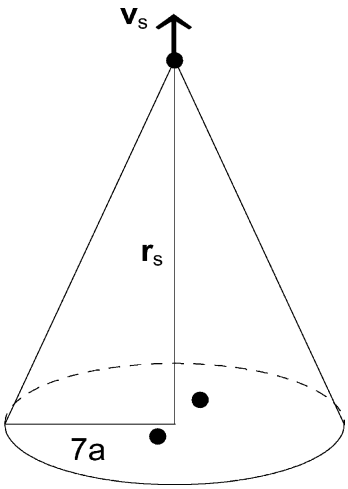


Figure 2. The loss cone. The escaping body, at the apex of the cone, must have come from inside the cone in order for a strong interaction to have occurred with the binary in the past. In reverse, bodies falling into the cone will interact efficiently with the binary and will be perturbed to new orbits. Thus, these orbits are 'lost'.

carry out some of the integrations. Following the steps of Monaghan (1976a), we get

$$\sigma \propto \int\int \frac{\mathrm{d}E_\mathrm{B}}{|E_\mathrm{B}|^{9/2}} e\,\mathrm{d}e. \tag{9}$$

The functions inside the integral give the distributions of binary energy:

$$f(|E_\mathrm{B}|)\mathrm{d}|E_\mathrm{B}| = 3.5|E_0|^{7/2}|E_\mathrm{B}|^{-9/2}\mathrm{d}|E_\mathrm{B}|, \tag{10}$$

and binary eccentricity:

$$f(e)\,\mathrm{d}e = 2e\,\mathrm{d}e \tag{11}$$

in a three-body break-up. The latter result has been obtained by Ambartsumian (1937) and Heggie (1975) using the principle of detailed balance in a stellar system.

The calculation may be repeated for planar systems with the result

$$f(|E_\mathrm{B}|)\mathrm{d}|E_\mathrm{B}| = 2|E_0|^2|E_\mathrm{B}|^{-3}\mathrm{d}|E_\mathrm{B}| \tag{12}$$

and

$$f(e)\,\mathrm{d}e = e(1-e^2)^{-1/2}\,\mathrm{d}e. \tag{13}$$

As far as the eccentricities go, these results are identical to those of Monaghan (1976a), but the energy distributions have steeper power laws. Numerical experiments have confirmed that indeed the present modification of the work of Monaghan (1976a) does represent an improvement. Monaghan's negative power-law index $n = 2$ is too shallow, while $n = 4.5$ of equation (10) agrees well with data.

4.2 Angular momentum dependence

Mikkola & Valtonen (1986) have shown that the power-law index is actually a function of the total angular momentum L_0. In general, we may write

$$f(|E_\mathrm{B}|)\mathrm{d}|E_\mathrm{B}| = (n-1)|E_0|^{n-1}|E_\mathrm{B}|^{-n}\mathrm{d}|E_\mathrm{B}|, \tag{14}$$

where

$$n - 3 = 18\,L^2. \tag{15}$$

The quantity L is the normalized total angular momentum, that is, L_0 divided by

$$L_\mathrm{max} = 2.5G\left(\frac{m_0^5}{|E_0|}\right)^{1/2}. \tag{16}$$

The value of L equals to $0.4\sqrt{w}$ and $n = 2.88w + 3$. Here, m_0 is the average mass of the three bodies defined [see Marchal et al. (1984)] by

$$m_0 = \sqrt{\frac{m_\mathrm{a}m_\mathrm{b} + m_\mathrm{a}m_\mathrm{s} + m_\mathrm{b}m_\mathrm{s}}{3}}. \tag{17}$$

We see that L goes from 0 to 1 while w goes from 0 to 6.25. To give an idea of the magnitude of L in a well-known three-body system, we note that for the Lagrangian equilateral equal-mass triangle with circular orbits we have approximately $L = 0.85$. This is definitely a system of relatively high angular momentum.

Even though the values of n in equation (15) are based on numerical data, we may justify them by the following arguments. When $L = 0$, the motion is restricted to a plane, and therefore we would expect the result derived for the planar non-rotating systems, $n = 3$. When the angular momentum L is very high, the vector sum $\boldsymbol{L}_0 = \boldsymbol{L}_\mathrm{B} + \boldsymbol{L}_\mathrm{s}$ of the angular momenta of the binary and the escaper should be large, as a rule. This is possible only if $\boldsymbol{L}_\mathrm{s}$ makes most of the contribution to $\boldsymbol{L}_0$ and $\boldsymbol{L}_\mathrm{B}$ is relatively small. The small values of $\beta = L_\mathrm{B}/L_\mathrm{s}$ at large w in Table 1 confirm this. The angular momentum of the binary L_B and its energy E_B are connected by

$$L_\mathrm{B}^2 = \mathcal{M}\frac{(Gm_\mathrm{a}m_\mathrm{b})^2}{2|E_\mathrm{B}|}(1-e^2). \tag{18}$$

Therefore, a small angular momentum generally (for a constant eccentricity) means that $|E_\mathrm{B}|$ has to be large. The probability that the binding energy is greater than $|E_\mathrm{B}|$ is proportional to $|E_\mathrm{B}|^{-7/2}$, according to equation (10). When equation (9) is multiplied by this factor we obtain approximately

$$\sigma \propto \int\int \frac{\mathrm{d}|E_\mathrm{B}|}{|E_\mathrm{B}|^8} e\,\mathrm{d}e. \tag{19}$$

Therefore, we would expect the power-law index $n = 8$ at high angular momentum when L_s is rather high. But as equation (15) shows, this is not the maximal value of n, but it becomes as great as $n = 16$ for the Lagrangian case, and possibly even greater when $L \to 1$.

4.3 Escape angle

The escape angle θ (see Fig. 3) is primarily controlled by the available phase space for different combinations of angular momenta. Therefore, we calculate the phase-space volume for an ensemble of escape orbits which have a common magnitude of the total angular momentum L_0.

The total angular momentum $\boldsymbol{L}_0$ is the vector sum of the binary angular momentum $\boldsymbol{L}_\mathrm{B}$ and the angular momentum of the escaping body $\boldsymbol{L}_\mathrm{s}$. Let the angle between $\boldsymbol{L}_\mathrm{s}$ and $\boldsymbol{L}_0$ be ϕ (Fig. 4). Then,

$$L_\mathrm{B}^2 = L_0^2 + L_\mathrm{s}^2 - 2L_0L_\mathrm{s}\cos\phi. \tag{20}$$

From this we find

$$L_0 = L_\mathrm{s}\cos\phi\left[1 \pm \sqrt{1 + \frac{\left(L_\mathrm{B}^2 - L_\mathrm{s}^2\right)}{L_\mathrm{s}^2\cos^2\phi}}\right]. \tag{21}$$

Since we deal only with positive L_0, the condition for choosing the minus sign in equation (21) is $L_\mathrm{B}^2 \leqslant L_\mathrm{s}^2$. In numerical simulations

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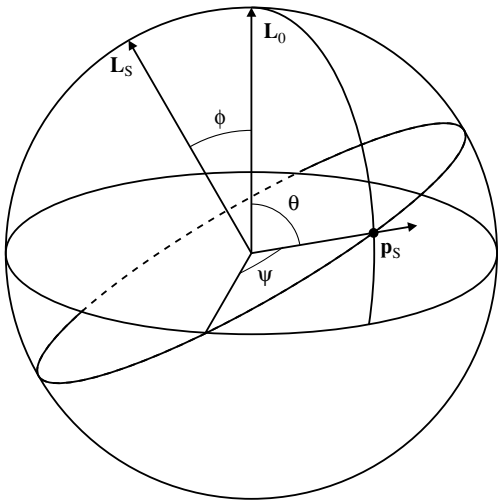


Figure 3. The geometry of the vectors $\boldsymbol{L}_0$ and $\boldsymbol{p}_s$ and the angles between them. The momentum vector $\boldsymbol{p}_s$ lies in a plane perpendicular to $\boldsymbol{L}_s$ while this plane is at an angle ϕ relative to the fundamental plane perpendicular to $\boldsymbol{L}_0$. The direction of $\boldsymbol{p}_s$ is specified by two angles θ and ψ.

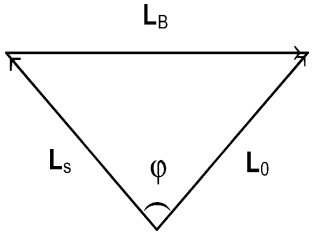


Figure 4. The relation between the vectors $\boldsymbol{L}_0$, $\boldsymbol{L}_s$ and $\boldsymbol{L}_B$. $\boldsymbol{L}_0$ is the vector sum of $\boldsymbol{L}_s$ and $\boldsymbol{L}_B$, and the angle between $\boldsymbol{L}_s$ and $\boldsymbol{L}_0$ is ϕ.

with a wide range of angular momenta (Valtonen et al. 2003), the average value of $L_B/L_s \approx 0.2$. Therefore, both signs are possible. The squares of the angular momenta are natural phase-space coordinates. Thus, we label $x = L_s^2$ and $y = L_B^2$.

The escape direction may be specified by angles ϕ and ψ (Fig. 3) whereby

$$\cos\theta = \sin\phi \sin\psi. \tag{22}$$

Let us first consider the special case of $\psi = \pi/2$. Then the momentum vector of the escaper $\boldsymbol{p}_s$ lies in the plane defined by vectors $\boldsymbol{L}_s$ and $\boldsymbol{L}_0$ and $\theta = \pi/2 - \phi$. The natural phase-space coordinate associated with ϕ is $\sin\phi = \cos\theta$ since $\cos\theta$ is uniformly distributed between -1 and $+1$ if the escape direction is random. We call this phase-space coordinate ζ: $1 - \zeta^2 = \cos^2\phi$. Using these natural coordinates,

$$L_0 = \sqrt{x}\sqrt{1-\zeta^2}\left\{1 \pm \sqrt{1 + \frac{(y-x)}{[x(1-\zeta^2)]}}\right\}. \tag{23}$$

The available phase-space volume may now be written as

$$\sigma = \int\int\int \delta(L^\star - L_0)\,\mathrm{d}x\,\mathrm{d}y\,\mathrm{d}\zeta, \tag{24}$$

where

$$L^\star = \sqrt{x}\sqrt{1-\zeta^2}\left\{1 \pm \sqrt{1 + \frac{(y-x)}{[x(1-\zeta^2)]}}\right\}.$$

The limits of integration are: x from 0 to ∞, ζ from 0 to 1, and y from 0 to a finite maximum value

$$y_{\max} = \frac{\mathcal{M}(Gm_a m_b)^2}{2|E_B|}. \tag{25}$$

The quantity inside the square root of equation (23) should not be negative. Thus

$$\zeta^2 \leqslant y/x. \tag{26}$$

This is a serious limitation to the available phase space. When $x \gg y$, $|\zeta|$ is confined to the neighbourhood of $|\zeta| \approx 0$. Therefore, escape directions tend to concentrate in the plane perpendicular to the angular momentum $\boldsymbol{L}_0$ (Saari 1974).

We may avoid the negative square root by introducing a new variable k in place of y by

$$y/x = (k^2+1)\zeta^2 \tag{27}$$

where k is a real number. Transforming to this variable we have

$$\sigma = \int\int\int \delta(L^\star - L_0) 2xk\zeta^2\,\mathrm{d}x\,\mathrm{d}k\,\mathrm{d}\zeta, \tag{28}$$

where

$$L^\star = \sqrt{x}(\sqrt{1-\zeta^2} \pm k\zeta).$$

The integrations over x and k may be carried out in a straightforward manner, with the result

$$\sigma = \int F(\zeta)\,\mathrm{d}\zeta, \tag{29}$$

where

$$F(\zeta) = \frac{2}{3}L_0^3 \frac{\sqrt{1-\zeta^2} \pm 3k_0\zeta}{\left(\sqrt{1-\zeta^2} \pm k_0\zeta\right)^3}. \tag{30}$$

Here k_0 is the lower limit of integration over k, the upper limit being ∞. The parameter k_0 is viewed here as a free parameter, to be determined on the basis of numerical experiments.

Let us consider both distributions $F(\zeta)$ corresponding to plus and minus signs. When $k_0 = \sqrt{1-\zeta^2}/\zeta = \cot\phi$, the distribution with minus sign goes to infinity and it is not suitable. For $k_0 \gg \cot\phi$, both distributions are in agreement. Therefore, we will use only the plus sign in the following.

So far we have dealt with the special case of $\theta = \pi/2 - \phi$. But in fact for every value of ϕ there is a whole range of escape directions from $\theta = \pi/2 - \phi$ to $\theta = \pi/2$ (here we neglect $\theta > \pi/2$ because of symmetry). Thus, the distribution of equation (30) is akin to the accumulated distribution of θ. In the first approximation, $F(\zeta_0)$ gives the relative weight of all escape orbits up to $\zeta \leqslant \zeta_0$. That is,

$$\frac{F(\zeta_0) - F(0)}{F(1) - F(0)} \approx \int_0^{\zeta_0} f(\zeta)\,\mathrm{d}\zeta. \tag{31}$$

Here $f(\zeta)$ is the distribution of the escape directions $\zeta = \cos\theta$.

In order to make $F(\zeta)$ in equation (30) a proper accumulated distribution, it has to be normalized. It can be done simply if we approximate $\zeta^2 \approx 0$ and consider $k_0 \gg 1$. Then the form of the accumulated distribution becomes

$$F(\zeta) = \frac{(1+k_0)^2}{k_0(k_0+2)}\left[1 - \frac{1}{(1+k_0\zeta)^2}\right] \tag{32}$$

and the corresponding distribution function becomes

$$f(\zeta) = \frac{(1+k_0)^2}{k_0+2}\frac{2}{(1+k_0\zeta)^3}. \tag{33}$$

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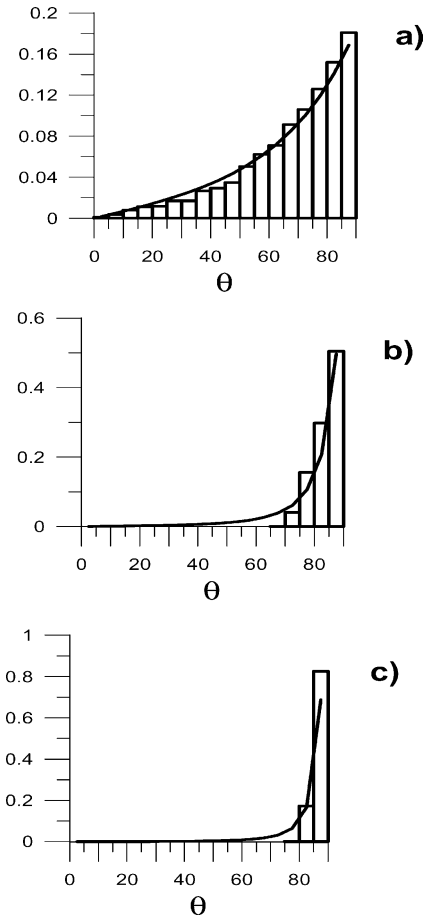


Figure 5. Comparison of distributions $f(\theta)$ for (a) $w = 0.1$; (b) $w = 2$ and (c) $w = 6$.

From numerical experiments we find that the parameter k_0 depends on the normalized angular momentum as

$$k_0 = 9L^{1.25}. \tag{34}$$

When $L \to 0$, the distribution $f(\zeta)$ is flat, as it should be since there cannot be any preferred escape direction relative to $\boldsymbol{L}_0 = 0$. The greater the value of L, the greater is the concentration towards $\theta = \pi/2$. In Figs 5(a)–(c), we compare numerical experiments with equation (33). The agreement is reasonable. The full phase-space calculation by Nash & Monaghan (1978) tends to give distributions of θ, which are too wide.

The angle between the vectors $\boldsymbol{L}_{\rm B}$ and $\boldsymbol{L}_{\rm s}$, λ, should follow sin (λ) if the two vectors are randomly oriented relative to each other. Figs 6(a) and (b) show that generally this is true if $w \approx 1$. But for lower w the distribution is skewed towards high values of λ, while at higher w the distribution has maximum weight at low λ.

4.4 Lifetime

The evolution of a three-body system takes place through periods of strong triple interaction, usually called interplay, and ejections when one of the three bodies recedes to some distance from the other two. We may define ejection such that the semimajor axis of the third-body orbit a_3 is greater than 10 times the semimajor axis a of the binary. With this definition, a triple system spends typically 1/3 of its lifetime in interplay and 2/3 in ejections (Anosova & Orlov 1986). In the following, we will estimate what is the typical time spent in ejections before the triple system breaks up, and then multiply the time by 1.5 to get an estimate for the total lifetime of the system.

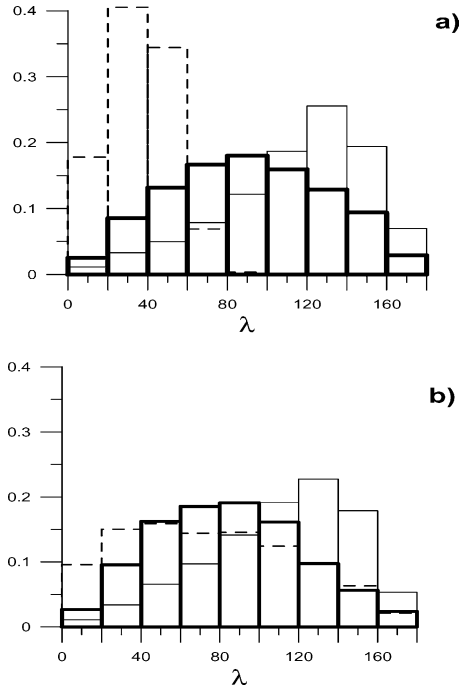


Figure 6. Distributions $f(\lambda)$ for $w = 0.1$ (solid lines), $w = 1$ (thick lines) and $w = 6$ (dashed lines). Fig. 6(a) corresponds to the first method of generating initial data (hierarchy) while Fig. 6(b) corresponds to the second method (region *D*).

The potential energy between the binary and the escaper forms a potential barrier which the escaper has to overcome. If it does not break through the barrier, the escaper returns, and we have effectively an ejection. Therefore, the escaper has to have excess energy $E_{\rm s}$ to truly escape.

Let us measure the energies in the inverse scale, normalized to the total energy of the system, and call this dimensionless quantity z. Thus for the binary of energy $E_{\rm B}$,

$$z = \frac{|E_0|}{|E_{\rm B}|}. \tag{35}$$

With this change of variables, equation (16) becomes

$$f(z) = (n-1)z^{n-2}. \tag{36}$$

We may write

$$z = \frac{|E_0|}{|E_{\rm B}|} = \frac{(|E_{\rm B}| - E_{\rm s})}{|E_{\rm B}|} = 1 - \frac{E_{\rm s}}{|E_{\rm B}|}, \tag{37}$$

which shows that extending the range of $E_{\rm s}$ to negative numbers, that is, to bound three-body orbits, means extending the range of z to values above unity. In the z-scale, we may associate a range $(1, 1 + {\rm d}z)$ to the potential barrier which the true escaper has to overcome. How wide should dz be? If the escaper has semimajor axis at least $10a$, it is easily estimated that ${\rm d}z \approx 0.25$.

The surface area of the distribution (36) between 1 and 1.25 may be referred to as capture zone. It represents the fraction $P_{\rm ej}$ of the total area under this distribution:

$$P_{\rm ej} = \frac{(1+{\rm d}z)^{n-1} - 1}{(1+{\rm d}z)^{n-1}} = 1 - 0.8^{n-1}. \tag{38}$$

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We call $P_{\rm ej}$ ejection probability. It ranges from $P_{\rm ej} = 0.36$ when $L = 0$ to $P_{\rm ej} \approx 0.75$ when $L = 0.5$. As an alternative to ejection, the escape has the probability of $1 - P_{\rm ej}$ per strong three-body encounter.

The probability that N ejections happen in succession is $P_{\rm ej}^N$. Therefore, the lifetime of the system exceeds N ejections with this probability. We want to find out how long a time-span these N ejections represent. Thus, we estimate a typical ejection time, and express it most conveniently in units of the crossing time (2). The orbital period of an ejection orbit is

$$T_{\rm orb} = 2\pi\sqrt{\frac{a_3^3}{GM}} \geqslant 2\pi 10^{3/2} G \left[\frac{(m_{\rm a}m_{\rm b})^3}{M}\right]^{1/2} (2|E_0|)^{-3/2}, \qquad (39)$$

where we have made use of equation (7) and inequality $a_3 > 10a$, as well as the fact that $E_{\rm B} \approx E_0$ for ejections. In terms of crossing times,

$$\frac{T_{\rm orb}}{T_{\rm cr}} \geqslant 2\pi 10^{3/2} \left(\frac{\sqrt{m_{\rm a}m_{\rm b}}}{M}\right)^3. \qquad (40)$$

For equal masses $T_{\rm orb}/T_{\rm cr}$ is greater than 7.36. Let us take z from the lower part of the capture zone, for example, $z = 1 + 0.25/3$. Then we get $T_{\rm orb} \approx 26T_{\rm cr}$ which in fact is what is found in numerical experiments (Anosova & Orlov 1986).

The half-life $T_{1/2}$ of a system is defined such that the probability of the system surviving this long as a bound triple is 50 per cent, that is, $P_{\rm ej}^N = 0.5$. From this we solve the number of ejections within half-life, $N = 0.3/\log(1/P_{\rm ej})$, and if each ejection lasts on average $26T_{\rm cr}$, $T_{1/2} = 7.8/\log(P_{\rm ej}^{-1})T_{\rm cr}$. The distribution of lifetimes T which follow from the process of repeated escape trials is an exponential (Valtonen & Aarseth 1977):

$$f(T) = \frac{0.69\,{\rm e}^{-0.69T/T_{1/2}}}{T_{1/2}}. \qquad (41)$$

The values of calculated half-lives $T_{1/2}$ for an equal-mass system are 12, 13, 29, 60, 233 and 830 $T_{\rm cr}$ for $w = 0.005$, 0.1, 1, 2, 4 and 6, respectively. These may be compared with the experimental values in Table 1. The agreement is generally good for $w = 0.1$, 1 and 2 and rather poor for $w = 4$ and 6. This may be connected with the fact that a significant fraction of triple systems does not break up during $1000T_{\rm cr}$ if w is large.

4.5 Eccentricity

The eccentricity distribution follows equation (11), but there may be angular momentum dependence also since the distribution for planar cases (which includes zero angular momentum) is different (equation 13) from the general case. Also at a very high angular momentum, equation (11) is not expected to hold (Standish 1972; Monaghan 1976b). We may use

$$f(e) = 2(p+1)e(1-e^2)^p \qquad (42)$$

for all angular momenta with the index

$$2p = L - 0.5. \qquad (43)$$

Figs 7(a) and (b) show that it works at the low end ($w = 0.1$) and high end ($w = 6$) of angular momenta for the H-cases. Considering all the experiments, we may also state that the distribution $f(e) = 2e$ covers well the entire range of parameters.

The median values $e_{1/2}$ for distribution (42) are 0.78, 0.76, 0.72, 0.70, 0.67 and 0.65 for $w = 0.005$, 0.1, 1, 2, 4 and 6, respectively. Those are in reasonable agreement with the experimental data in Table 1.

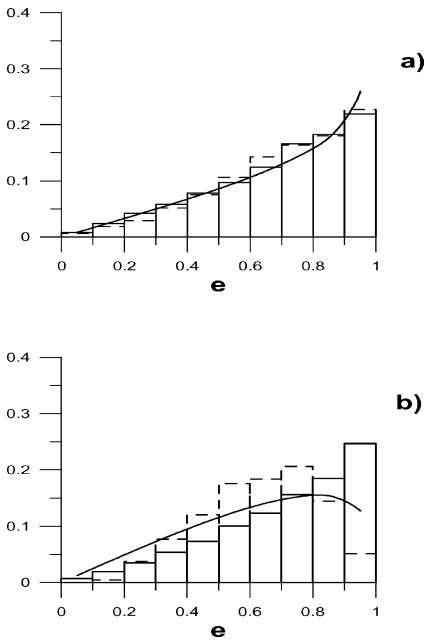


Figure 7. Comparison of distributions $f(e)$ for (a) $w = 0.1$ and (b) $w = 6$. The histograms for hierarchical data are shown by dashed lines and for the region D – solid lines. The curves show the fitting by equation (42).

4.6 Escape velocity

The escape velocity of the third body $v_{\rm s}$ in the centre-of-mass coordinate system is given by

$$f(v_{\rm s})\,{\rm d}v_{\rm s} \propto \frac{v_{\rm s}\,{\rm d}v_{\rm s}}{\left[|E_0| + \frac{1}{2}(m_{\rm s}M/m_{\rm B})v_{\rm s}^2\right]^{9/2}}. \qquad (44)$$

For a general value of angular momentum and equal masses, this equation becomes

$$f(v_{\rm s}){\rm d}v_{\rm s} = \frac{3}{2}(n-1)|E_0|^{n-1}v_{\rm s}\left(|E_0| + \frac{3}{4}v_{\rm s}^2\right)^{-n}{\rm d}v_{\rm s}. \qquad (45)$$

The median value of the distribution is

$$(v_{\rm s})_{\rm med} = \frac{2\left[2^{1/(n-1)} - 1\right]^{1/2}|E_0|^{1/2}}{3^{1/2}}. \qquad (46)$$

For our calculated values of $w = 0.005$, 0.1, 1, 2, 4 and 6, $(v_{\rm s})_{\rm med} =$ 1.35, 1.26, 0.83, 0.65, 0.49 and 0.41, respectively. These values are in reasonable agreement with the medians found from simulations for the H cases. The trends are less clear in the D cases (see Table 1).

Fig. 8 compares the theoretical distribution with numerical experiments. At high angular momenta, the H-type experiments and the theory give distributions peaked at lower velocities (Fig. 8c). In contrast, the theory of Nash & Monaghan (1978) leads to the opposite expectation, high velocities at the high values of their angular momentum parameter j_0. The distribution of Nash & Monaghan (1978) at low angular momentum $j_0 = 0.25$, roughly corresponding to $w = 0.1$, also peaks at higher $v_{\rm s}$ than our experimental distributions.

4.7 Semimajor axis

The distribution of the semimajor axes of the remaining binaries is closely related to the escape velocity distribution. The median values $a_{1/2}$ for $w = 0.1$, 1, 2, 4 and 6 are expected to be

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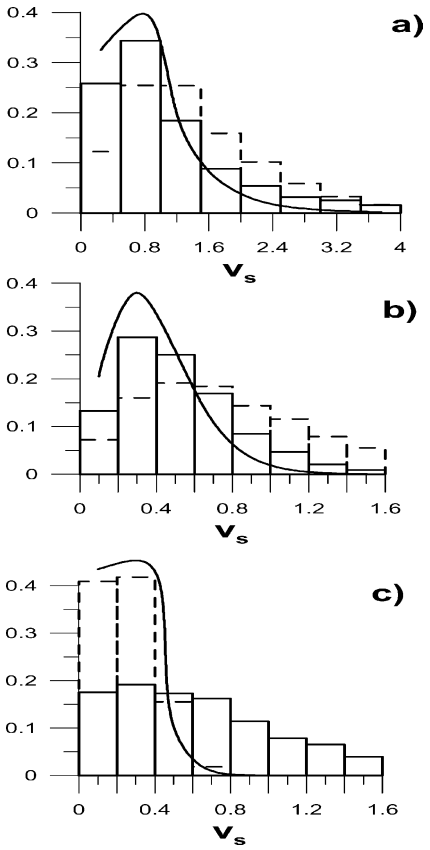


Figure 8. Comparison of escape-velocity distributions for (a) $w = 0.1$; (b) $w = 2$; and (c) $w = 6$. The curves correspond to fitting by equation (45).

$a_{1/2} = 0.11$, 0.12, 0.13, 0.15 and 0.16, respectively, again in agreement with Table 1.

Sterzik and Durisen (1998) considered the case of zero angular momentum ($w = 0$). Our results (semimajor axis and escaper velocity) for $w = 0.005$ are in agreement with their results.

4.8 Angular momenta

The ratio of the binary angular momentum to the escaper angular momentum β has to be $\beta = 1$ when $L_0 = 0$, because then the two angular momenta are equal in magnitude but opposite in direction. When $L \to 1$, the contribution of L_B becomes small in comparison to L_s. Since the loss cone of escape orbits extends to about $7a$, at the limit the escaper angular momentum L_s should be about seven times the binary angular momentum L_B, that is, $\beta \approx 1/7$. This agrees with Table 1.

5 CONCLUSIONS

Here, we have demonstrated that several quantities describing the three-body break-up may be derived from the ergodic principle, with the main limitations arising from the constants of motion. The total angular momentum turns out to be the crucial parameter in describing the distributions. This work is an extension of the theory of Monaghan and associates, and its general principles are confirmed. The later work of Mikkola and Valtonen is also confirmed and it is extended to cover a wide range of escape distributions. Therefore, it appears that the three-body problem has analytical solutions even for strongly interacting systems, but these solutions are necessarily of statistical nature. In many astrophysical applications, the exact initial conditions for the three bodies are not known, or even required, and statistical distributions are exactly what is needed (Heggie 1975).

The differences between the two different samples, D and H, are largely explained by the short-lived and initially unstable systems. For systems with $T > 10\,T_{cr}$ and $w = 6$ (column 8 in Table 1), the statistical outcomes are practically identical for both the samples, which enforces the ergodic principle.

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REFERENCES

Aarseth S. J., Zare K., 1974, Celest Mech, 10, 185
Agekian T. A., Anosova J. P., 1967, Sov. Astron., 44, 1261
Ambartsumian V. A., 1937, Sov. Astron., 14, 207
Anosova J. P., 1969, Tr. Astron. Obs. Leningr. Univ., 26, 88
Anosova J. P., Orlov V. V., 1985, Tr. Astron. Obs. Leningr. Univ., 40, 66
Anosova J. P., Orlov V. V., 1986, Sov. Astron., 63, 643
Anosova J. P., Bertov D. I., Orlov V. V., 1984, Astrofis., 20, 177
Bulirsch R., Stoer J., 1966, Num. Math., 8, 1
Heggie D. C., 1975, MNRAS, 173, 729
Marchal C., Yoshida J., Sun Y.-S., 1984, Celest. Mech., 34, 65
Mikkola S., Aarseth S. J., 1993, Celest. Mech. Dyn. Astron., 57, 439
Mikkola S., Valtonen M., 1986, MNRAS, 223, 269
Monaghan J. J., 1976a, MNRAS, 176, 63
Monaghan J. J., 1976b, MNRAS, 177, 583
Nash P. E., Monaghan J. J., 1978, MNRAS, 184, 119
Saari D. G., 1974, Celest. Mech., 9, 175
Saslaw W. C., Valtonen M. J., Aarseth S. J., 1974, ApJ, 190, 253
Standish E. M., 1972, A&A, 21, 185
Sterzik M. F., Durisen R. H., 1998, A&A, 339, 95
Valtonen M., 1974, in Kozai Y., ed., The Stability of the Solar System and of Small Stellar Systems. D. Reidel, Dordrecht, p. 211
Valtonen M., 1988, Vistas Astron., 32, 23
Valtonen M., Aarseth S. J., 1977, Rev. Mex. Astron. Astrofis., 3, 163
Valtonen M. J., Karttunen H., 2005, The Three-Body Problem. Cambridge Univ. Press, Cambridge
Valtonen M., Mikkola S., 1991, ARA&A, 29, 9
Valtonen M., Mylläri A. A., Orlov V. V., Rubinov A. V., 2003, Astron. Lett., 29, 41

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